Solution

(1) (15 pts.) Is A invertible? If true, find A^{-1}.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{array}\right]$$

$$\begin{array}{l} R_1 - R_2 \\ \\ R_3 + R_2 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 1 & 1 \end{array}\right] \quad \begin{array}{l} R_1 - R_3 \\ -R_2 \\ \\ \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & -2 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 & 1 & 1 \end{array}\right]$$

Yes

$$A^{-1} = \begin{bmatrix} -2 & -1 & -2 \\ -1 & 0 & -1 \\ 2 & 1 & 1 \end{bmatrix} \qquad \det A = 1$$

(2) (10 pts.) Find the determinant of the matrix A.

$$A = \begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

use row operations

$$\xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_4 - R_3} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

expand cofactors R_1

$$\det A = -\begin{vmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix} - \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= -(1) - (1) = -2$$

$$\det A = (-2)(-1)(-1)$$

$$\det A = -2$$

(3) (25 pts.) Answer the following for the matrix A and vectors $\mathbf{v}$, $\mathbf{w}$:

$$A = \begin{bmatrix} 1 & 0 & 0 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & -1 & 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

(a) Describe all solutions for $A\mathbf{x} = \mathbf{v}$ in parametric vector form.
(b) Find the rank and nullity of A.
(c) Find a basis for the column space of A.
(d) Find a basis for the null space of A.
(e) True or False: $\mathbf{v}$ is in the column space of A.
(f) True or False: $\mathbf{w}$ is in the null space of A.

augmented matrix $[A \mid \bar{v}] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 6 & 1 \\ 0 & 1 & 2 & 3 & -2 \\ 0 & 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_3-R_2]{} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 6 & 1 \\ 0 & 1 & 2 & 3 & -2 \\ 0 & 0 & -3 & -3 & 3 \end{array}\right]$

12

$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 6 & 1 \\ 0 & 1 & 2 & 3 & -2 \\ 0 & 0 & 1 & 1 & -1 \end{array}\right] \xrightarrow[R_2-2R_3]{} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 6 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 \end{array}\right]$

$-\frac{1}{3}R_3$

7 (a) $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1-6x_4 \\ -x_4 \\ -1-x_4 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -6 \\ -1 \\ -1 \\ 1 \end{pmatrix}$

2 (b) Rank $A = 3$, Nullity $A = 1$

1 (c) C_1, C_2, C_3 of A: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$

1 (d) Null space $= x_4 \begin{pmatrix} -6 \\ -1 \\ -1 \\ 1 \end{pmatrix}$ Basis: $\begin{pmatrix} -6 \\ -1 \\ -1 \\ 1 \end{pmatrix}$

1 (e) True $A\bar{x} = \bar{v}$ is consistent.

1 (f) $A\bar{w} = \begin{pmatrix} -5 \\ 0 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ False

(4) (25 pts.) Answer the following for the set W and vector $\mathbf{w}$:

$$W = \left\{ \begin{bmatrix} b+c+d \\ a+b \\ c+d \\ a+b+c \end{bmatrix} : a, b, c, d, \in \mathbb{R} \right\}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

(a) Find a matrix A such that W is the column space of A.
(b) True or False: $\mathbf{w}$ is in W. If true, solve for $\mathbf{x}$: $A\mathbf{x} = \mathbf{w}$.
(c) Find the determinant, rank and nullity of A.
(d) True or False: W is equal to $\mathbb{R}^4$.

augmented matrix $= [A \mid w] = \left[\begin{array}{cccc|c} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow[R_4-R_2]{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array}\right]$

$R_1 - R_2 \left[\begin{array}{cccc|c} 1 & 0 & -1 & -1 & -1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array}\right]$ $\begin{array}{l} R_1+R_3 \\ R_2-R_3 \\ \\ R_4-R_3 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right]$ $\begin{array}{l} R_3+R_4 \\ -R_4 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$

(a) $A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ 4

(b) True 1 $\quad \bar{x} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$ 4

5 (c) $\det A = (-1)(-1) = 1$
1 rank $A = 4$
1 null $A = 0$

1 (d) True

(5) (25 pts.) Answer the following for the matrix A and vector $\mathbf{w}$:

$$A = \begin{bmatrix} 1 & -1 & 0 & -3 \\ 1 & 0 & 2 & -3 \\ 0 & 1 & -2 & 5 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ -4 \\ 3 \\ 2 \end{bmatrix}$$

(a) Find a basis for the column space of A.
(b) Find a basis for the null space of A.
(c) Find the determinant, rank and nullity of A.
(d) True or False: $\mathbf{w}$ is in the column space of A.
(e) True or False: A is invertible.
(f) True or False: $\mathbf{w}$ is in the null space of A.

augmented matrix $[A|w] = \left[\begin{array}{cccc|c} 1 & -1 & 0 & -3 & 2 \\ 1 & 0 & 2 & -3 & -4 \\ 0 & 1 & -2 & 5 & 3 \\ -1 & 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow[R_4+R_1]{R_2-R_1} \left[\begin{array}{cccc|c} 1 & -1 & 0 & -3 & 2 \\ 0 & 1 & 2 & 0 & -6 \\ 0 & 1 & -2 & 5 & 3 \\ 0 & -1 & 0 & -2 & 4 \end{array}\right]$

$\begin{array}{c} R_1+R_2 \\ \\ R_3-R_2 \\ R_4+R_2 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 2 & -3 & -4 \\ 0 & 1 & 2 & 0 & -6 \\ 0 & 0 & -4 & 5 & 9 \\ 0 & 0 & 2 & -2 & -2 \end{array}\right] \begin{array}{c} R_1-R_4 \\ R_2-R_4 \\ R_3+2R_4 \\ \frac{1}{2}R_4 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -2 \\ 0 & 1 & 0 & 2 & -4 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 1 & -1 & -1 \end{array}\right] \xrightarrow{R_3 \leftrightarrow R_4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -2 \\ 0 & 1 & 0 & 2 & -4 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$

5

$\begin{array}{c} R_1+R_4 \\ R_2-2R_4 \\ R_3+R_4 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -14 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$

3 (a) c_1, c_2, c_3, c_4 of A

(b) no basis for Null Sp A

5 (c) $\det A = (-1)(2) = -2$ 2 Rank$(A) = 4$, 2 Null$(A) = 0$

3 (d) True $\bar{w} = A \begin{bmatrix} 3 \\ -14 \\ 4 \\ 5 \end{bmatrix}$

3 (e) True

2 (f) False